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PIEZOELECTRIC VIBRATOR

Abstract

A piezoelectric vibrator including: a vibrator layer having a vibrating portion, a holding portion surrounding a periphery of the vibrating portion, and a holding arm connecting the holding portion and the vibrating portion, the vibrating portion having a piezoelectric layer, a first electrode layer, and a second electrode layer; a first lid layer on a first electrode layer side of the vibrator layer; a second lid layer on a second electrode layer side of the vibrator layer; a first joining member joining the holding portion of the vibrator layer and the first lid layer; a second joining member joining the holding portion of the vibrator layer and the second lid layer; and a covering member covering an entire periphery of at least one of an inner surface and an outer surface of at least one of the first joining member and the second joining member.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of International application No. PCT/JP2023/024586, filed Jul. 3, 2023, which claims priority to Japanese Patent Application No. 2022-209035, filed Dec. 26, 2022, the entire contents of each of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a piezoelectric vibrator.

BACKGROUND ART

[0003] In recent years, the miniaturization of vibrators has advanced. In general, a crystal vibrator with a wafer-level package structure obtained by bonding three wafers together is known, wherein the three wafers are a lid layer, a vibrator layer and a substrate layer. For example, Patent Document 1 discloses a crystal vibrator obtained by bonding three crystal wafers together using a low-melting glass or polyimide resin, which is an insulating material. Further, Patent Document 2 discloses a crystal vibrator obtained by joining three crystal wafers using Au, which is a conductor.

[0004] Patent Document 1: Japanese Unexamined Patent Application Publication No. 2013-38524

[0005] Patent Document 2: Japanese Unexamined Patent Application Publication No. 2022-99603

SUMMARY OF THE DISCLOSURE

[0006] However, in the configuration of Patent Document 1, when the three crystal wafers are bonded together using a resin, since the resin generally has a high gas permeability, moisture and oxygen may enter the crystal vibrator from a resin sealing portion formed by the resin. Thus, moisture and oxygen entering the crystal vibrator react with the metal of the electrode film of a crystal vibrating element, causing the mass of the crystal vibrating element to fluctuate; and as a result, the long-term frequency stability of the crystal vibrating element may be impaired. Also, when each of the joining members of the three wafers is made of Au metal, as described in Patent Document 2, it is necessary to make the surface roughness of each surface of the wafers extremely small, so that it is difficult to reduce the cost.

[0007] The present disclosure has been made in view of the above problems, and an object of the present disclosure is to provide a piezoelectric vibrator with high reliability at low cost.

[0008] A piezoelectric vibrator according to an aspect of the present disclosure includes: a vibrator layer having a vibrating portion, a holding portion surrounding a periphery of the vibrating portion in a plan view of the piezoelectric vibrator, and a holding arm connecting the holding portion and the vibrating portion, the vibrating portion having a piezoelectric layer, first electrode layer on a first principal surface of the piezoelectric layer, and a second electrode layer on a second principal surface of the piezoelectric layer, the second principal surface facing the first principal surface; a first lid layer on a first electrode layer side of the vibrator layer; a second lid layer provided on a second electrode layer side of the vibrator layer; a first joining member surrounding the vibrating portion in the plan view and joining the holding portion of the vibrator layer and the first lid layer; a second joining member surrounding the vibrating portion in the plan view and joining the holding portion of the vibrator layer and the second lid layer; and a covering member covering an entire periphery of at least one of an inner surface and an outer surface of at least one of the first joining member and the second joining member, wherein the at least one of the first joining member and the second joining member is made of a first insulating material, and the covering member is made of a second insulating material having a gas permeability lower than that of the first insulating

material.

[0009] In the present disclosure, the covering member made of the second insulating material having lower gas permeability than the first insulating material is used. By covering the entire periphery of at least one of the inner surface and the outer surface of at least one of the first joining member and the second joining member with the covering member, the gas-tight sealability of the interior can be enhanced, so that the frequency fluctuation of the piezoelectric vibrator can be suppressed. Further, since at least one of the first joining member and the second joining member is made of the first insulating material, the cost can be reduced compared with a configuration in which all joining members are made of metal. Therefore, it is possible to provide a piezoelectric vibrator with high reliability corresponding to reduced cost.

[0010] According to the present disclosure, it is possible to provide a piezoelectric vibrator with high reliability and at low cost.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is an exploded perspective view schematically showing a piezoelectric vibrator according to a first embodiment.

[0012] FIG. 2 is a cross-sectional view of the piezoelectric vibrator according to the first embodiment taken along line II-II of FIG. 1.

[0013] FIG. 3 is a cross-sectional view showing a configuration of a piezoelectric vibrator according to a second embodiment.

[0014] FIG. 4 is a cross-sectional view showing a configuration of a piezoelectric vibrator according to a third embodiment.

[0015] FIG. 5 is a plan view showing a configuration of a crystal vibrating element according to a fourth embodiment.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0016] Embodiments of the present disclosure will be described below with reference to the drawings. Note that the drawings of the present embodiment are exemplary, the dimensions and shapes of each component are schematic, and the technical scope of the present disclosure should not be construed as limiting to the present embodiment.

[0017] For convenience, each drawing may be accompanied by a Cartesian coordinate system consisting of X axis, Y' axis and Z' axis to clarify the relationship between each drawing and to help understand the positional relationship of each component. The X axis, Y' axis and Z' axis correspond to each other in each drawing. The X axis, Y' axis and Z' axis correspond to the crystallographic axes of a quartz crystal blank, which is to be described later, respectively. The X axis corresponds to the electrical axis (polarity axis) of the quartz crystal, the Y axis corresponds to the mechanical axis of the quartz crystal, and the Z axis corresponds to the optical axis of the quartz crystal. The Y' axis and the Z' axis are axes obtained by rotating the Y axis and the Z axis by 35 degrees 15 minutes $\pm$ 1 minute 30 seconds from the Y axis to the Z axis around the X axis, respectively.

First Embodiment

[0018] A configuration of a piezoelectric vibrator **1** according to a first embodiment of the present disclosure will be described with reference to FIGS. 1 and 2. FIG. 1 is an exploded perspective view schematically showing the piezoelectric vibrator **1** according to the present embodiment. FIG. 2 is a cross-sectional view of the piezoelectric vibrator **1** according to the present embodiment taken along line II-II of FIG. 1.

[0019] The piezoelectric vibrator **1** according to the present embodiment includes a crystal vibrating element **10**, a first joining member **20**, an upper lid layer **30**, a second joining member **40**,

and a lower lid layer **50**. In the following description, the crystal vibrating element **10** having a quartz crystal blank **12** will be described as an example of a piezoelectric vibrating element of the piezoelectric vibrator **1**. The quartz crystal blank **12** is a kind of piezoelectric body (piezoelectric piece) that vibrates in response to a voltage applied thereto. Note that the piezoelectric vibrating element is not limited to the crystal vibrating element **10**, but other piezoelectric bodies such as ceramic or the like may be used as the piezoelectric vibrating element. Further, the piezoelectric vibrating element may alternatively be a MEMS vibrating element manufactured using MEMS technology.

[0020] The crystal vibrating element **10** (corresponding to an example of a “vibrator layer”) is an element that vibrates a quartz crystal by a piezoelectric effect to convert electric energy and mechanical energy. The crystal vibrating element **10** includes an AT-cut type quartz crystal blank **12**. When axes obtained by rotating, among an X axis, a Y axis, and a Z axis as crystallographic axes of synthetic quartz crystal, the Y axis and the Z axis about the X axis by 35 degrees 15 minutes ± 1 minute 30 seconds in the direction from the Y axis to the Z axis are defined as a Y' axis and a Z' axis, respectively, the AT-cut type quartz crystal blank **12** is cut out such that its principal surface lies in an XZ' plane specified by the X axis and the Z' axis. The quartz crystal blank **12** may be a tuning fork type quartz crystal blank.

[0021] Note that the rotation angle of the Y' axis and the Z' axis of the AT-cut type quartz crystal blank **12** may be tilted from 35 degrees 15 minutes by an amount within a range of -5 degrees to $+15$ degrees. Alternatively, cuts other than AT cut, such as BT cut, GT cut, SC cut, and the like may be used as the cut angle of the quartz crystal blank **12**.

[0022] A crystal vibrating element using the AT-cut type quartz crystal blank has high frequency stability over a wide temperature range. Further, an AT-cut crystal vibrating element has excellent aging characteristics and can be manufactured at low cost. Further, the AT-cut crystal vibrating element uses a thickness shear vibration mode as the main vibration.

[0023] The crystal vibrating element **10** includes a set of excitation electrodes. An alternating electric field is applied between the set of excitation electrodes. With such a configuration, the vibrating portion of the quartz crystal blank **12** vibrates at a predetermined oscillation frequency by the thickness shear vibration mode, so that resonance characteristics associated with the vibration are obtained.

[0024] Thus, since the main vibration of the crystal vibrating element **10** is the thickness shear vibration mode, a crystal vibrating element that performs thickness shear vibration at a vibration frequency in an MHz band can be easily realized by using, for example, the AT-cut type quartz crystal blank **12**.

[0025] The quartz crystal blank **12** (corresponding to an example of a “piezoelectric layer”) has a first principal surface **12a** and a second principal surface **12b** lying in the XZ' plane and facing each other. The quartz crystal blank **12** has a flat plate shape. Therefore, the first principal surface **12a** and the second principal surface **12b** of the quartz crystal blank **12** are flat surfaces, respectively. Note that the quartz crystal blank **12** is not limited to a flat plate shape, but may have a convex or concave central portion, for example.

[0026] The AT-cut type quartz crystal blank **12** has a long side direction in which a long side parallel to the X axis direction extends, a short side direction in which a short side parallel to the Z' axis direction extends, and a thickness direction in which a thickness parallel to the Y' axis direction extends. The quartz crystal blank **12** has a rectangular shape when the first principal surface **12a** of the quartz crystal blank **12** is viewed in plan view (hereinafter simply referred to as “in plan view”). The quartz crystal blank **12** may be thinned to a predetermined thickness by, for example, polishing after being joined to the second joining member **40** in a state having a larger thickness.

[0027] The planar shape of the quartz crystal blank **12** is not limited to a rectangular shape. The planar shape of the quartz crystal blank **12** may alternatively be a polygonal shape, a circular shape,

an elliptical shape, or a combination of these shapes.

[0028] The crystal vibrating element **10** includes a vibrating portion **16**, a holding arm **17**, and a holding portion **18**. The vibrating portion **16** has the quartz crystal blank **12** and the set of excitation electrodes. The set of excitation electrodes includes a first excitation electrode **14a** (corresponding to an example of an “upper electrode layer”) and a second excitation electrode **14b** (corresponding to an example of a “lower electrode layer”). The first excitation electrode **14a** is provided on the first principal surface **12a** of the vibrating portion **16**, and the second excitation electrode **14b** is provided on the second principal surface **12b** of the vibrating portion **16**. The first excitation electrode **14a** and the second excitation electrode **14b** are provided facing each other with the quartz crystal blank **12** sandwiched therebetween. The first excitation electrode **14a** and the second excitation electrode **14b** each have a rectangular shape when the first principal surface **12a** is viewed in plan view, and are arranged so that substantially the whole of the first excitation electrode **14a** and substantially the whole of the second excitation electrode **14b** overlap each other in the XZ' plane.

[0029] The shape of each of the first excitation electrode **14a** and the second excitation electrode **14b** is not limited to a rectangular shape, but may be a polygonal shape, a circular shape, an elliptical shape, or a combination of these shapes.

[0030] The crystal vibrating element **10** has extended electrodes and connecting electrodes (both not shown), and the first excitation electrode **14a** is electrically connected to an outer electrode **51a**, and the second excitation electrode **14b** is electrically connected to an outer electrode **51d**, for example, by these electrodes. By applying an alternating electric field to the first excitation electrode **14a** and the second excitation electrode **14b** via the outer electrodes **51a** and **51d**, the vibrating portion **16** (specifically, a portion of the quartz crystal blank **12** where the first excitation electrode **14a** and the second excitation electrode **14b** are provided) vibrates in a predetermined vibration mode.

[0031] The materials of the first excitation electrode **14a**, the second excitation electrode **14b**, the extended electrodes, and the connecting electrodes are, for example, aluminum (Al), molybdenum (Mo), or gold (Au). Each of the electrodes described above may be, for example, a multilayer body composed of a titanium (Ti) layer provided on the quartz crystal blank **12** side and a gold (Au) layer provided on the surface side.

[0032] The crystal vibrating element **10** is accommodated in an inner space **60** formed between the first joining member **20**, the upper lid layer **30**, the second joining member **40**, and the lower lid layer **50**. The inner space **60** is gas-tightly sealed, for example. Note that the inner space **60** may be gas-tightly sealed in a vacuum state or gas-tightly sealed in a state filled with a gas such as an inert gas.

[0033] The vibrating portion **16** is a portion of the crystal vibrating element **10** and is located in a central portion of the inner space **60**. As shown in FIG. 2, the vibrating portion **16** has the first principal surface **12a** and the second principal surface **12b** facing the first principal surface **12a**. The vibrating portion **16** has the first excitation electrode **14a** provided on the first principal surface **12a** and the second excitation electrode **14b** provided on the second principal surface **12b**. The first excitation electrode **14a** is provided on a surface of the quartz crystal blank **12** facing the upper lid layer **30**, and the second excitation electrode **14b** is provided on a surface of the quartz crystal blank **12** facing the lower lid layer **50**.

[0034] Similar to the vibrating portion **16**, the holding arm **17** is also located in the inner space **60**, and connects the vibrating portion **16** and the holding portion **18**. The extended electrodes (not shown) are formed on the holding arm **17**, and can lead the first excitation electrode **14a** and the second excitation electrode **14b** to the connecting electrodes provided on the holding portion **18**.

[0035] The holding portion **18** is formed in a frame shape that surrounds the periphery of the vibrating portion **16** in plan view. The holding portion **18** is joined to the first joining member **20** and the second joining member **40** in the vertical direction. The holding portion **18** is connected to

the holding arm **17**.

[0036] A covering member **19** is made of an insulating material having lower gas permeability than an insulating material of the second joining member **40**, which will be described later. For example, aluminum oxide (Al_2O_3) is used as the insulating material of the covering member **19**. In the example shown in FIG. 2, the covering member **19** is provided to cover an inner surface **40a** of the second joining member **40**. Specifically, the covering member **19** is provided on the surface of the crystal vibrating element **10**, the inner surface **40a** of the second joining member **40**, and an upper surface of the lower lid layer **50** facing the vibrating portion. For example, in a case where a first wafer constituting a plurality of lower lid layers **50**, a second wafer constituting a plurality of crystal vibrating elements **10**, and a wafer constituting a plurality of upper lid layers **30** are bonded to each other, and then each wafer is cut with a dicing machine to obtain a plurality of piezoelectric vibrators **1**, the covering member **19** can be formed after the first wafer and the second wafer are bonded to each other. The thickness of the covering member **19** is 1 nm to 100 nm, and preferably 5 nm to 50 nm. ALD (atomic layer deposition), CVD (chemical vapor deposition), PVD (physical vapor deposition), or the like can be used as the film formation method of the covering member **19**. A film with low gas permeability can be easily formed by the ALD, CVD, or PVD described above. In particular, when the ALD is used, the covering member **19** with low gas permeability can be formed even in a thick structure with uneven surfaces and undulations. The covering member **19** is not limited to being provided in the locations described above, but may be provided, for example, only on the inner surface **40a** of the second joining member **40**, as in the second embodiment to be described later, or on an outer surface **20b** of the first joining member **20** and an outer surface **40b** of the second joining member **40**, as in a third embodiment to be described later.

[0037] In the present embodiment, the first joining member **20** is made of a metallic material. The first joining member **20** is provided between an upper surface side of the holding portion **18** and the upper lid layer **30** to metal-join the holding portion **18** and the upper lid layer **30**. The first joining member **20** is provided so as to surround the vibrating portion **16** in plan view, and joins the holding portion **18** of the crystal vibrating element **10** and the upper lid layer **30** in a frame shape. The first joining member **20** has the inner surface **20a** and the outer surface **20b**, wherein the inner surface **20a** is in contact with the inner space **60**, and the outer surface **20b** is on the opposite side of the inner surface **20a** with the first joining member **20** sandwiched therebetween. Au—Au joining, AuSn joining, AlGe joining, or the like may be used as the metal joining as long as airtightness can be obtained in the inner space **60**.

[0038] The upper lid layer **30** is formed in a flat plate shape, for example. The dimensions of the upper lid layer **30** in plan view are the same or substantially the same as the dimensions of the crystal vibrating element **10**. The upper lid layer **30** is provided on the first excitation electrode **14a** side of the crystal vibrating element **10**. The upper lid layer **30** is joined to the first joining member **20** to thereby form a portion of the inner space **60** by. The upper lid layer **30** may be made of quartz crystal, or, alternatively, may be made of silicon, ceramic, or glass.

[0039] In the present embodiment, the second joining member **40** is made of an insulating material. For example, a resin or a glass may be used as the insulating material of the second joining member **40**. For example, polyimide resin may be used as the resin. Also, a low-melting glass may be used as the glass. The second joining member **40** is provided between the lower surface side of the holding portion **18** and the lower lid layer **50** to join the holding portion **18** and the lower lid layer **50**. The second joining member **40** is provided so as to surround the vibrating portion **16** in plan view, and joins the holding portion **18** of the crystal vibrating element **10** and the lower lid layer **50** in a frame shape. The second joining member **40** has the inner surface **40a** and the outer surface **40b**, wherein the inner surface **40a** faces the inner space **60**, and the outer surface **40b** is on the opposite side of the inner surface **40a** with the second joining member **40** sandwiched therebetween. In the present embodiment, the covering member **19** is provided on the inner surface **40a**. With such a configuration, the gas permeating through the second joining member **40** is

blocked by the covering member **19**, so that the gas-tight sealability of the inner space **60** can be improved. Further, by forming the second joining member **40** on the side of the lower lid layer **50** where outer electrodes **51**, which will be described later, are provided, with an insulating material, it is possible to achieve miniaturization and good electrical characteristics compared to, for example, the configuration disclosed in Patent Document 2, in which a metal joining is used. That is, in the configuration disclosed in Patent Document 2 in which the metal joining is used, since wiring lines cannot be drawn across the joining member made of metal, it is necessary to form via(s) in the vibrating element, so that it is difficult to secure a sufficient area of the vibrating portion. On the other hand, in the configuration of the present embodiment, since the crystal vibrating element **10** and the lower lid layer **50** are joined by the second joining member **40** made of an insulating material, the wiring lines can be drawn across the second joining member **40**, so that it is not necessary to form the via(s) in the crystal vibrating element **10**. Therefore, it is possible to maximize the first excitation electrode **14a** and the second excitation electrode **14b** of the vibrating portion **16** to obtain good electrical characteristics equivalent to those of a large-sized product and to achieve miniaturization of the piezoelectric vibrator **1**.

[0040] The lower lid layer **50** is formed in a flat plate shape, for example. The dimensions of the lower lid layer **50** in plan view are the same or substantially the same as the dimensions of the crystal vibrating element **10**. The lower lid layer **50** is provided on the second excitation electrode **14b** side of the crystal vibrating element **10**. The lower lid layer **50** is joined to the second joining member **40** to thereby form a portion of the inner space **60**. The lower lid layer **50** may be made of quartz crystal, or, alternatively, may be made of silicon, ceramic, or glass. By forming at least one of the upper lid layer **30** and the lower lid layer **50** with silicon, glass, or ceramic having low transparency to visible light, image recognition, mounting positioning and the like can be easily performed when mounting the piezoelectric vibrator **1**. The lower lid layer **50** has the outer electrodes **51** on the side opposite to the side facing the crystal vibrating element **10**.

[0041] Outer electrodes **51a**, **51b**, **51c**, and **51d** are provided at four corners of the lower lid layer **50**. The outer electrode **51a** is electrically connected to the first excitation electrode **14a**, and the outer electrode **51d** is electrically connected to the second excitation electrode **14b**. The outer electrodes **51b** and **51c** may be used as dummy electrodes that are not electrically connected to the crystal vibrating element **10**. With such an electrode arrangement, when mounting the piezoelectric vibrator **1** on a substrate, even if the piezoelectric vibrator **1** is mounted upside down, the piezoelectric vibrator **1** can be mounted because the positions of the outer electrodes are symmetrical. Further, by securing the maximum distance between the outer electrode **51a** and the outer electrode **51d**, a short circuit between the two electrodes can be suppressed.

[0042] The inner space **60** is formed inside the piezoelectric vibrator **1** by the first joining member **20**, the upper lid layer **30**, the second joining member **40**, and the lower lid layer **50**. The vibrating portion **16** and the holding arm **17** are provided in the inner space **60** to constitute a vibration space of the crystal vibrating element **10**.

[0043] As described above, in the piezoelectric vibrator **1** according to the present embodiment, the inner surface **40a** of the second joining member **40** is covered by the covering member **19**. With such a configuration, the gas entering from the outside of the piezoelectric vibrator **1** through the second joining member **40** is blocked by the covering member **19** made of an insulating material having a gas permeability lower than that of the insulating material of the second joining member **40**, so that the gas-tight sealability of the inner space **60** can be enhanced. Further, by forming the second joining member **40** with an insulating material, the cost can be reduced as compared with a configuration in which all joining members are made of metal.

[0044] In the present embodiment, the covering member **19** is provided on the inner surface **40a** of the second joining member **40**; however, the covering member **19** is not limited to being provided on the inner surface **40a** of the second joining member **40**, but may be provided to cover the entire periphery of at least one of the inner surfaces **20a** and **40a** and the outer surfaces **20b** and **40b** of at

least one of the first joining member **20** and the second joining member **40**. For example, the covering member **19** may be provided on the outer surface **40b** of the second joining member **40** or on both the inner surface **40a** and the outer surface **40b** of the second joining member **40**. The present embodiment describes an aspect in which the crystal vibrating element **10** and the upper lid layer **30** are metal-joined by the first joining member **20**; however, when the first joining member **20** is made of an insulating material, at least one of the inner surface **20a** and the outer surface **20b** of the first joining member **20** may be covered by the covering member **19**, similar to the case of the second joining member **40**.

[0045] Further, the present embodiment describes a configuration in which the first joining member **20** is made of a metallic material, the second joining member **40** is made of an insulating material, and the second joining member **40** is covered by the covering member **19**; however, the present disclosure is not limited to such a configuration as long as at least one of the joining members is made of an insulating material. For example, when the first joining member **20** is made of an insulating material, the first joining member **20** may be covered by the covering member **19**.

[0046] Further, in the present embodiment, the covering member **19** is provided on surfaces of each of the first excitation electrode **14a** and the second excitation electrode **14b** of the crystal vibrating element **10**. With such a configuration, by covering the surfaces of the first excitation electrode **14a** and the second excitation electrode **14b** with the covering member **19**, oxidation of the first excitation electrode **14a** and the second excitation electrode **14b** can be suppressed.

[0047] A configuration of a resin sealing device and a resin sealing method according to another embodiment of the present disclosure will be described below. In the following embodiments, descriptions of the matters common to those of the first embodiment will be omitted and only the differences will be described. In particular, similar effects due to similar configurations will not be described one by one.

Second Embodiment

[0048] Next, the structure of a piezoelectric vibrator **2** according to a second embodiment will be described with reference to FIG. **3**. FIG. **3** is a cross-sectional view schematically showing the structure of the piezoelectric vibrator according to the second embodiment.

[0049] The present embodiment differs from the first embodiment in the positions where the covering member **19** is provided. Specifically, in the first embodiment, the covering member **19** is provided on the surface of the crystal vibrating element **10**, the inner surface **40a** of the second joining member **40**, and the upper surface of the lower lid layer **50** facing the vibrating portion; while in the present embodiment, a covering member **19a** is partially provided on the inner surface **40a** of the second joining member **40**. In such a case, the covering member **19a** is provided to avoid a surface of the crystal vibrating element **10** joined to the first joining member **20**. With such a configuration, the joinability between the crystal vibrating element **10** and the upper lid layer **30** can be improved. Further, the covering member **19a** is provided to avoid each surface of the first excitation electrode **14a** and the second excitation electrode **14b** of the vibrating portion **16**. With such a configuration, when the covering member **19a** has the gas adsorption property, the vibration characteristic fluctuation of the vibrating portion **16** due to the weight fluctuation of the covering member can be suppressed.

Third Embodiment

[0050] Next, the structure of a piezoelectric vibrator **3** according to a third embodiment will be described with reference to FIG. **4**. FIG. **4** is a cross-sectional view schematically showing the structure of the piezoelectric vibrator according to the third embodiment.

[0051] The present embodiment differs from the first embodiment in the positions where the covering member **19** is provided. Specifically, in the first embodiment, the covering member **19** is provided on the surface of the crystal vibrating element **10**, the inner surface **40a** of the second joining member **40**, and the upper surface of the lower lid layer **50** facing the vibrating portion; while in the present embodiment, a covering member **19b** is provided on the entire side surface of

the piezoelectric vibrator **3** so as to cover the outer surface **20b** of the first joining member **20** and the outer surface **40b** of the second joining member **40**. With such a configuration, since not only the second joining member **40** but also the first joining member **20** can be covered, the first joining member can be formed an insulating material, so that the cost can be further reduced.

Fourth Embodiment

[0052] Next, the structure of a crystal vibrating element **110** according to a fourth embodiment will be described with reference to FIG. 5. FIG. 5 is a plan view schematically showing the structure of the crystal vibrating element according to the fourth embodiment.

[0053] In the present embodiment, unlike the first embodiment, the crystal vibrating element **110** further includes functional element(s) that assist the function of the piezoelectric vibrator. For example, as shown in FIG. 5, a temperature sensor mounting portion **121** and a temperature sensor **122** are provided in the inner space **60**. The crystal vibrating element **110** has a holding arm **117a** that connects a vibrating portion **116** and a holding portion **118**, and a holding arm **117b** that connects the holding portion **118** and the temperature sensor mounting portion **121**. The temperature sensor **122** is provided on the temperature sensor mounting portion **121**. For example, a thermistor that detects a change in temperature as a change in resistance value is used as the temperature sensor **122**. By providing the thermistor, frequency temperature correction can be improved as compared with a conventional crystal vibrator. When a thermistor is used, the temperature change can be detected more accurately by disposing the thermistor near a quartz crystal blank **112** and a first excitation electrode **114a** of the vibrating portion **116** as shown in FIG. 5. Further, the temperature of the quartz crystal blank **112** can be detected more accurately by performing adjustment so that the heat flow rate from the substrate on which the crystal vibrator is equal between the vibrating portion **116** and the temperature sensor mounting portion **121**, and performing adjustment so that the heat capacity of the quartz crystal blank **112** and the heat capacity of the temperature sensor **122** are equal.

[0054] An inductor may be used as an example of the functional element(s). The inductor is composed as a portion of an LC circuit and has a function as a high pass filter that cuts frequencies lower than a predetermined frequency. By providing the inductor on the crystal vibrating element **110**, it is possible to provide a crystal vibrating element with an inductor that can be manufactured at a lower cost than a configuration in which a filter using an inductor is formed on an external circuit. Further, by providing the functional element(s) in the inner space **60** with a high gas-tight sealability, deterioration of the functional element(s) over time can be suppressed. Further, by providing the functional element(s) in the inner space **60**, it is possible to reduce the number of protective components of the functional element(s).

[0055] A part or all of the embodiments of the present disclosure will be described below. The present disclosure is not limited to the following description.

[0056] <1> A piezoelectric vibrator according to an aspect of the present disclosure comprising: a vibrator layer having a vibrating portion, a holding portion surrounding a periphery of the vibrating portion in plan view, and a holding arm connecting the holding portion and the vibrating portion, the vibrating portion having a piezoelectric layer, an upper electrode layer provided on a first principal surface of the piezoelectric layer, and a lower electrode layer provided on a second principal surface of the piezoelectric layer, the second principal surface facing the first principal surface; an upper lid layer provided on an upper electrode layer side of the vibrator layer; a lower lid layer provided on a lower electrode layer side of the vibrator layer; a first joining member provided so as to surround the vibrating portion in plan view and joining the holding portion of the vibrator layer and the upper lid layer; a second joining member provided so as to surround the vibrating portion in plan view and joining the holding portion of the vibrator layer and the lower lid layer; and a covering member provided so as to cover an entire periphery of at least one of an inner surface and an outer surface of at least one of the first joining member and the second joining member, wherein the at least one of the first joining member and the second joining member is

made of a first insulating material, and the covering member is made of a second insulating material having a gas permeability lower than that of the first insulating material.

[0057] According to the above aspect, by covering the inner surface and outer surface of at least one of the first joining member and the second joining member with a covering member having low gas permeability, gas entering from the first joining member or the second joining member can be suppressed, and long-term frequency stability can be improved. Further, by forming at least one of the first joining member and the second joining member with the first insulating material, it is possible to reduce the cost as compared with a configuration in which all joining members are made of metal.

[0058] <2> As an aspect, the piezoelectric vibrator according to <1>, wherein the lower lid layer has an outer electrode on a side opposite to a side facing the vibrator layer, the first joining member is made of a metallic material, and the second joining member is made of the first insulating material.

[0059] According to the above aspect, compared with a configuration in which the second joining member and the lower lid layer are joined by metal, it is not necessary to provide via(s) in the lower lid layer, so that the upper electrode layer and the lower electrode layer can be maximized, which corresponds to miniaturization.

[0060] <3> As an aspect, the piezoelectric vibrator according to <2>, wherein the vibrator layer further has a functional element.

[0061] <4> As an aspect, the piezoelectric vibrator according to <1>, wherein the covering member is provided to avoid each surface of the upper electrode layer and the lower electrode layer of the vibrating portion.

[0062] According to the above aspect, it is possible to prevent the frequency characteristics of the vibrating portion from fluctuating due to the gas adsorption property of the covering member.

[0063] <5> As an aspect, the piezoelectric vibrator according to any one of <1> to <3>, wherein the covering member is further provided on each surface of the upper electrode layer and the lower electrode layer of the vibrating portion.

[0064] According to the above aspect, by covering the upper electrode and the lower electrode with the covering member, the frequency fluctuation due to the oxidation of the electrodes can be suppressed.

[0065] <6> As an aspect, the piezoelectric vibrator according to any one of <1> to <3>, wherein the covering member is provided to avoid a surface of the vibrator layer joined to the first joining member.

[0066] According to the above aspect, the joinability between the vibrator layer and the first joining member can be improved.

[0067] <7> As an aspect, the piezoelectric vibrator according to any one of <1> to <6>, wherein the piezoelectric layer of the vibrator layer is made of quartz crystal.

[0068] <8> As an aspect, the piezoelectric vibrator according to any one of <1> to <7>, wherein the upper lid layer and the lower lid layer are made of quartz crystal.

[0069] <9> As an aspect, the piezoelectric vibrator according to any one of <1> to <7>, wherein at least one of the upper lid layer and the lower lid layer is made of silicon, ceramic, or glass.

[0070] According to the above aspect, image recognition can be easily performed when mounting the piezoelectric vibrator to a substrate.

[0071] <10> As an aspect, the piezoelectric vibrator according to any one of <1> to <9>, wherein the covering member has a thickness of 1 nm to 100 nm.

[0072] <11> As an aspect, the piezoelectric vibrator according to any one of <1> to <10>, wherein the first insulating material is made of resin or glass, and the second insulating material is made of aluminum oxide.

[0073] It should be noted that the embodiments described above are intended to facilitate understanding of the present disclosure, and are not intended to limit the interpretation of the

present disclosure. The present disclosure may be changed/improved without departing from its scope, and the present disclosure also includes equivalents thereof. In other words, appropriate design changes made to each embodiment by those skilled in the art are included in the scope of the disclosure as long as the features of the present disclosure are provided. For example, the respective elements included in each embodiment, and the arrangements, materials, conditions, shapes, sizes, and the like of the elements are not limited to those exemplified but can be changed appropriately. Further, the respective elements included in each embodiment may be combined as long as it is technically possible, and the combinations of the elements are within the scope of the present disclosure as long as they have the features of the present disclosure.

REFERENCE SIGNS LIST

[0074] **1, 2, 3** piezoelectric vibrator [0075] **10, 110** crystal vibrating element [0076] **12, 112** quartz crystal blank [0077] **12a, 112a** first principal surface [0078] **12b** second principal surface [0079] **14a, 114a** first excitation electrode [0080] **14b** second excitation electrode [0081] **16, 116** vibrating portion [0082] **17, 117a, 117b** holding arm [0083] **18, 118** holding portion [0084] **19, 19a, 19b** covering member [0085] **20** first joining member [0086] **20a** inner surface [0087] **20b** outer surface [0088] **30** upper lid layer [0089] **40** second joining member [0090] **40a** inner surface [0091] **40b** outer surface [0092] **50** lower lid layer [0093] **51, 51a, 51b, 51c, 51d** outer electrode [0094] **60** inner space [0095] **121** temperature sensor mounting portion [0096] **122** temperature sensor

Claims

1. A piezoelectric vibrator comprising: a vibrator layer having a vibrating portion, a holding portion surrounding a periphery of the vibrating portion in a plan view of the piezoelectric vibrator, and a holding arm connecting the holding portion and the vibrating portion, the vibrating portion having a piezoelectric layer, first electrode layer on a first principal surface of the piezoelectric layer, and a second electrode layer on a second principal surface of the piezoelectric layer, the second principal surface facing the first principal surface; a first lid layer on a first electrode layer side of the vibrator layer; a second lid layer provided on a second electrode layer side of the vibrator layer; a first joining member surrounding the vibrating portion in the plan view and joining the holding portion of the vibrator layer and the first lid layer; a second joining member surrounding the vibrating portion in the plan view and joining the holding portion of the vibrator layer and the second lid layer; and a covering member covering an entire periphery of at least one of an inner surface and an outer surface of at least one of the first joining member and the second joining member, wherein the at least one of the first joining member and the second joining member is made of a first insulating material, and the covering member is made of a second insulating material having a gas permeability lower than that of the first insulating material.
2. The piezoelectric vibrator according to claim 1, wherein the second lid layer has an electrode on a side thereof opposite to a side facing the vibrator layer, the first joining member is made of a metallic material, and the second joining member is made of the first insulating material.
3. The piezoelectric vibrator according to claim 2, wherein the vibrator layer further has a functional element.
4. The piezoelectric vibrator according to claim 1, wherein the covering member does not extend onto surfaces of each of the first electrode layer and the second electrode layer of the vibrating portion.
5. The piezoelectric vibrator according to claim 1, wherein the covering member is on surfaces of each of the first electrode layer and the second electrode layer of the vibrating portion.
6. The piezoelectric vibrator according to claim 1, wherein the covering member does not extend onto a surface of the vibrator layer joined to the first joining member.
7. The piezoelectric vibrator according to claim 1, wherein the piezoelectric layer of the vibrator layer comprises a quartz crystal.

- 8.** The piezoelectric vibrator according to claim 1, wherein the first lid layer and the second lid layer comprise a quartz crystal.
- 9.** The piezoelectric vibrator according to claim 1, wherein at least one of the first lid layer and the second lid layer comprises silicon, ceramic, or glass.
- 10.** The piezoelectric vibrator according to claim 1, wherein the covering member has a thickness of 1 nm to 100 nm.
- 11.** The piezoelectric vibrator according to claim 1, wherein the first insulating material comprises resin or glass, and the second insulating material comprises aluminum oxide.
- 12.** The piezoelectric vibrator according to claim 1, wherein the covering member covers the entire periphery of the inner surface of the second joining member, and the second joining member is made of the first insulating material.
- 13.** The piezoelectric vibrator according to claim 12, wherein the covering member is on the surface of the crystal vibrating element, and a surface of the second lid layer facing the vibrating portion.
- 14.** The piezoelectric vibrator according to claim 13, further comprising an electrode on a side of the second lid layer opposite to a side thereof facing the vibrator layer.
- 15.** The piezoelectric vibrator according to claim 13, wherein the covering member is on surfaces of each of the first electrode layer and the second electrode layer of the vibrating portion.
- 16.** The piezoelectric vibrator according to claim 1, wherein the covering member covers an entire periphery of the outer surfaces of both of the first joining member and the second joining member.
- 17.** The piezoelectric vibrator according to claim 16, wherein the covering member does not extend onto surfaces of each of the first electrode layer and the second electrode layer of the vibrating portion.
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